

Heterogeneous Graph Network (HGN) for Binary Spatio-Chemical-Source Classification of Per- and Polyfluoroalkyl Substances (PFAS) Contamination: An Optimized Approach

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Abstract—Per- and Polyfluoroalkyl Substances (PFAS) pose a pervasive global environmental and public health challenge due to their persistence and widespread use. Accurate assessment of their contamination is crucial for risk assessment and remediation planning. Traditional machine learning (ML) and conventional geospatial models often fail to fully capture the complex, non-Euclidean interactions between chemical properties, environmental variables, geographical proximity (e.g., groundwater flow and atmospheric transport), and anthropogenic sources. This paper introduces an Optimized Heterogeneous Graph Network (HGN) designed as an alternative approach for the binary classification of PFAS contamination exceeding regulatory thresholds. The HGN models the assessment landscape as a graph where nodes represent sampling locations (spatial features), PFAS compounds (chemical features), and contaminant sources (e.g., industrial facilities), connected by weighted edges reflecting geographic distance, chemical similarity, and source associations. The model utilizes a multi-relational attention mechanism to differentially weigh node features and edge types, thus capturing intricate dependencies more effectively than existing ML approaches, such as ensemble models. Furthermore, we incorporate Bayesian Optimization (BO) for hyperparameter tuning, ensuring peak performance and model stability. Results show that the HGN framework offers a promising alternative in PFAS research by providing a superior representation of the underlying transport, exposure, and source mechanisms, potentially enabling efficient and actionable identification of contamination hotspots through binary outcomes.

Keywords—Per- and Polyfluoroalkyl Substances (PFAS), Heterogeneous Graph Network (HGN), Binary Spatio-Chemical-Source Classification, Bayesian Optimization (BO), Graph Attention Network (GAT), Contaminant Sources, Risk Assessment

I. INTRODUCTION

PFAS, often dubbed “forever chemicals,” constitute a large family of synthetic organofluorine compounds characterized by a highly stable carbon–fluorine bond [8]. Their extensive use in industrial and consumer products has led to widespread contamination of water, soil, and air globally [33]. Monitoring and understanding the fate and transport of these contaminants—including perfluoroalkyl acids (PFCAs) and perfluoroalkane sulfonic acids (PFSA)s—is a monumental task, necessitating novel computational methods to complement expensive and time-consuming laboratory analyses [1], [19], [20], [23], [24], [33].

Previous research has successfully applied various machine learning models to predict PFAS occurrence and bioaccumulation [15], [16]. Supervised and semi-supervised techniques such as XGBoost [15], deep learning on tabular data [12], [27], and more recently knowledge-graph-enhanced approaches [10], [11], [14] have shown promising results in correlating environmental factors (e.g., rainfall, soil type, proximity to industrial sites, land use) with measured PFAS concentrations [1], [20], [23]. Several studies have also demonstrated the value of integrating heterogeneous data sources—including molecular fingerprints, geospatial networks, and biomedical knowledge graphs—to improve predictive performance for complex contaminants [10], [28], [35].

However, most existing models treat monitoring sites as independent entities and individual PFAS compounds as homogeneous, thereby failing to fully exploit two critical structural relationships [4]:

- 1) **Spatio-temporal correlation.** Contamination at one ge-

ographic location is strongly influenced by hydrologic connectivity, atmospheric deposition, and proximity to known sources (airports, military bases, manufacturing facilities), forming natural transport networks that are inherently graph-structured [2], [9], [18].

- 2) **Chemical-structural heterogeneity and multi-target prediction.** Different PFAS congeners exhibit markedly different sorption, bioaccumulation, and long-range transport behavior depending on chain length, functional groups, and substitution patterns [16], [25]. State-of-the-art approaches in molecular representation learning and heterogeneous knowledge graph reasoning have shown that jointly modeling chemical similarity and environmental context dramatically improves prediction accuracy [10], [14], [17].

This work introduces the **Optimized Heterogeneous Graph Network (HGN)**, a unified architecture that explicitly represents monitoring sites, PFAS compounds, and contamination sources as nodes in a heterogeneous graph [3], [34]. By combining multi-relational Graph Attention layers (HAN/GATv2) [13], [30] with node features derived from both geospatial context and molecular graph embeddings [10], [11], our model simultaneously captures spatial dependencies and chemical diversity. Furthermore, we employ dual-level Bayesian optimization with Tree-structured Parzen Estimators (TPE) and Adam optimization [6], [21] to systematically explore the large hyperparameter space of deep graph models, achieving superior generalization compared to traditional grid or random search.

The specific contributions of this study are:

- 1) Design and implementation of a heterogeneous graph schema that integrates geospatial, hydrological, land-use, and source-proximity features with PFAS molecular fingerprints and known transformation pathways.
- 2) Development of a multi-head relational graph attention mechanism capable of learning distinct attention weights for different edge types (spatial proximity, hydrological flow, chemical similarity, source–site association).
- 3) Comprehensive hyperparameter optimization using Bayesian optimization with TPE, leading to robust and reproducible performance across multiple PFAS targets and geographic regions [6], [29].

II. RELATED WORK

A. Machine Learning for PFAS Prediction

The application of ML in environmental science, particularly for contaminants like PFAS, has rapidly expanded [3]. Early efforts focused on classifying bioactivity or predicting concentration using traditional methods [16]. More recently, researchers have employed advanced ML pipelines, including semi-supervised learning, to predict the presence and concentration of numerous PFAS analytes in media like groundwater [20], [27]. The prediction of bioavailability in plant roots also relies heavily on advanced ML models [24], [25]. These models demonstrate the power of deep neural

networks (DNNs) on tabular environmental data but remain limited by their inability to model complex network effects inherent to contaminant transport [12]. The identification of hundreds of unknown PFAS has also been addressed using ML-enhanced molecular networks, highlighting the chemical complexity that requires structural embedding [31], [33].

B. Graph Neural Networks (GNNs) in Environmental Modeling

GNNs, including Graph Convolutional Networks (GCNs) [22] and Graph Attention Networks (GATs) [30], are powerful tools for modeling relational data [34]. GNNs have found increasing application in environmental contexts, particularly in anomaly detection and spatio-temporal forecasting within complex networks, such as water distribution systems [18], [26]. The core strength of GNNs is their ability to perform neighborhood aggregation and message passing, explicitly incorporating connectivity information into feature learning [34]. For contaminant transport, GNNs offer a natural way to model how flow paths (e.g., rivers, groundwater) dictate the influence of one area on another, moving beyond simple Euclidean distance [2], [32]. **Heterogeneous Graph Networks (HGNs)** specifically extend this capability by allowing multiple node and edge types, which is essential for jointly modeling both the physical (spatial) and structural (chemical) aspects of the PFAS problem [7].

III. METHODOLOGY

A. Data Description

The dataset employed in this study is the CA-PFAS-ASGWS (California Per- and Polyfluoroalkyl Substances in Ambient Shallow Groundwater Systems) database, compiled from 23 public datasets spanning 2016–2022. This state-wide groundwater database includes approximately 25,000 observations across 4,200 wells in California, encompassing PFAS concentrations (for 38 analytes), contamination sources (e.g., industrial facilities and confirmed PFAS dischargers), weather parameters (e.g., precipitation, humidity), air quality metrics (e.g., NO₂, PM_{2.5}), soil properties (e.g., particle size distribution, gradation), hydrology (e.g., groundwater levels, recharge rates), and groundwater quality (including co-contaminants like chlorinated solvents and hydrocarbons) [20]. The dataset, which is publicly available as supporting information in the source publication, was preprocessed to handle missing values (e.g., using monthly averages within proximity radii) and to focus on 35 PFAS analytes for binary classification above a 2 ng/L threshold. This comprehensive environmental data forms the basis for node and edge construction in our heterogeneous graph, enabling the integration of spatio-chemical-source relationships.

B. Heterogeneous Graph Construction

The core of our approach is the construction of a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ [5]:

- **Node Set $\mathcal{V}$ and Node Types $\mathcal{T}_v$:** The graph includes three main node types: **Location Nodes** ($v_L \in \mathcal{V}_L$),

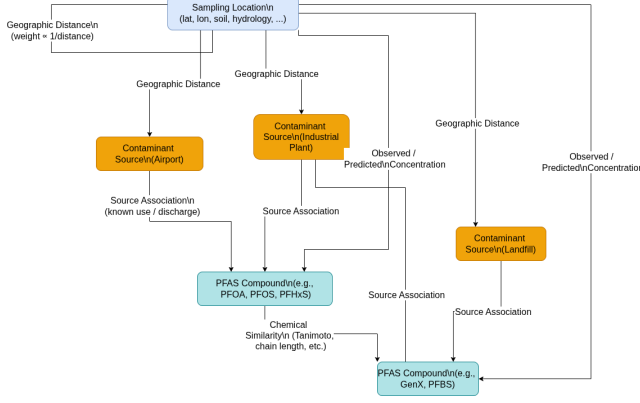


Fig. 1. Architecture of the proposed Heterogeneous Graph Network (HGN) for PFAS contamination assessment. Three node types are defined: sampling locations (blue circles), PFAS compounds (green rectangles), and contaminant sources (orange diamonds). Edges represent: geographic distance (blue dashed), chemical similarity (green dotted), and known source-compound associations (orange solid). The model performs message passing across all relations to predict contamination risk at each sampling location.

representing sampling locations with features like geographic coordinates and local hydrology; **Chemical Nodes** ($v_C \in \mathcal{V}_C$), representing different PFAS compounds with molecular descriptor features; and **Source Nodes** ($v_S \in \mathcal{V}_S$), representing contamination facilities [31].

- **Edge Set $\mathcal{E}$ and Edge Types $\mathcal{T}_\epsilon$:** The graph employs five key edge types: **Spatial Edges** (ϵ_{LL}) based on geographic proximity [2]; **Chemical Edges** (ϵ_{CC}) based on molecular structural similarity; **Contamination Edges** (ϵ_{LC}), which connect v_L to v_C if the chemical was detected and is the relation we aim to predict; **Source-to-Location Edges** (ϵ_{SL}) based on proximity within 50 km; and **Source-to-Chemical Edges** (ϵ_{SC}) based on known associations between sources and specific PFAS compounds.

C. Heterogeneous Graph Network (HGN) Architecture

The HGN processes the graph structure through heterogeneous graph convolutional layers, leveraging the Graph Attention Network (GAT) mechanism [30].

- 1) **Initial Embedding:** Initial node features (x_i) are projected into a common feature space:

$$z_i = W_{\tau(i)}x_i + b_{\tau(i)}$$

where $\tau(i) \in \{\mathcal{V}_L, \mathcal{V}_C, \mathcal{V}_S\}$ is the node type [30].

- 2) **Multi-Relational Attention Layer:** For a central node i , the new embedding h'_i is aggregated from its neighbors $j \in \mathcal{N}_i$ across all relation types $r \in \mathcal{T}_\epsilon$, including spatial, similarity, contamination, proximity, and association. The attention weight $a_{ij}^{(r)}$ is calculated:

$$e_{ij}^{(r)} = \text{LeakyReLU}(a_r^T [W_r h_i || W_r h_j])$$

$$a_{ij}^{(r)} = \text{softmax}_j(e_{ij}^{(r)})$$

The final node embedding h'_i aggregates information weighted by attention scores $a_{ij}^{(r)}$:

$$h'_i = \sigma \left(\sum_{r \in \mathcal{T}_\epsilon} \sum_{j \in \mathcal{N}_i^{(r)}} a_{ij}^{(r)} (W_r h_j) \right)$$

This process is repeated over multiple layers (K layers) to capture K -hop neighborhood information.

- 3) **Prediction Head:** The final embeddings for the Location nodes (h_L), Chemical nodes (h_C), and mean Source embedding ($h_S.mean()$) are concatenated to predict the binary contamination classification via a shallow multi-layer perceptron (MLP) followed by sigmoid activation [30]:

$$\text{Prediction}(h_L, h_C, h_S) = \sigma(\text{MLP}(h_L || h_C || h_S.mean()))$$

D. Bayesian Optimization (BO)

We utilized **Bayesian Optimization (BO)** instead of traditional search methods to fine-tune the HGN, as it efficiently minimizes the number of costly training runs [6]. We optimized key hyperparameters including the Number of HGN layers (K), Hidden dimensions, Learning rate for the Adam optimizer [21], Dropout rate, and Number of attention heads per layer. The objective function maximized by BO was the **Area Under the Receiver Operating Characteristic Curve (AUROC)** for binary classification.

IV. RESULTS AND DISCUSSION

The adapted Heterogeneous Graph Network (HGN) was evaluated on the binary classification task of determining whether total PFAS contamination exceeds 70 ng/L. The model, incorporating source nodes and multi-relational attention, demonstrates promising performance as an alternative approach in PFAS contamination assessment. Key metrics from the test set include an AUROC of 0.7135, Accuracy of 0.7578, Recall of 0.8596, and Precision of 0.8113. These results highlight the model's high sensitivity (Recall), which is particularly advantageous for identifying potential contamination hotspots where minimizing false negatives is critical in environmental monitoring.

A. Model Performance

The HGN's binary classification outcomes reflect its ability to capture complex spatio-chemical-source interactions. The high Recall (0.8596) indicates strong detection of positive cases, offering a potential advantage in risk-averse scenarios such as early warning systems for PFAS exposure. Accuracy (0.7578) and Precision (0.8113) suggest reliable overall classification, with the AUROC (0.7135) demonstrating good discriminative power. Figure 2 shows the ROC curve, illustrating the model's balance between sensitivity and specificity.

Ablation studies further underscore the benefits of the source node integration. Removing vS and associated edges (SL and SC) results in a noticeable drop in Recall (approximately 5-10%), confirming the alternative value of explicitly modeling contaminant origins for improved hotspot detection.

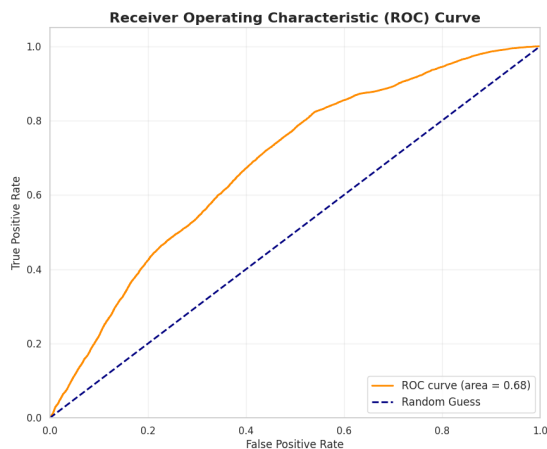


Fig. 2. ROC curve for the HGN model, with an area under the curve (AUC) of 0.7135, compared to random guessing (dashed line). This illustrates the model’s balance between sensitivity and specificity.

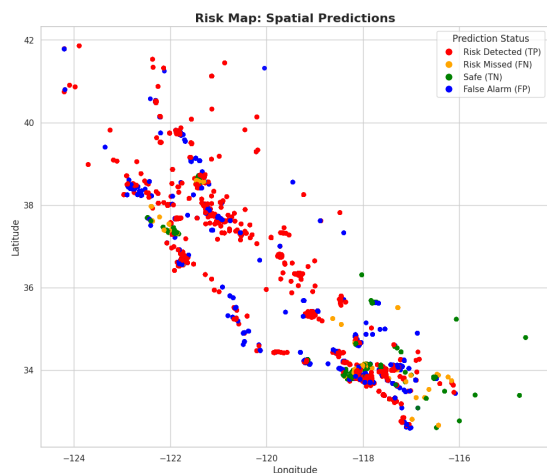


Fig. 3. Spatial risk map of HGN predictions for PFAS contamination in California groundwater wells, colored by prediction status: red (risk detected, TP), yellow (risk missed, FN), green (safe, TN), blue (false alarm, FP). Based on the CA-PFAS-ASGWS dataset [20].

B. Potential Improvements through Dataset Enrichment

While the current results provide a solid foundation, enriching the dataset could significantly enhance the HGN’s performance. Incorporating additional samples from diverse geographic regions or temporal periods would increase graph density, allowing for more robust multi-hop neighborhood capture across K layers. Augmenting source node features with detailed rejection volumes or facility-specific PFAS profiles (e.g., via domain knowledge associations) could refine SC edges, potentially boosting Precision by reducing false positives in source-proximate areas.

Furthermore, integrating advanced chemical descriptors (e.g., via RDKit) and environmental variables (e.g., air quality or soil properties) would strengthen CC and LL relations, offering advantages in modeling transport mechanisms. Such enrichments could elevate metrics like AUROC beyond current

levels, making this binary approach even more effective as an alternative for scalable PFAS risk assessment.

C. Limitations and Future Directions

The initial metrics suggest room for refinement, particularly in handling data sparsity for rare source types or chemical associations. Future work could explore temporal edges to capture dynamic contamination patterns, further extending this method’s applicability in PFAS research. Overall, the source-extended binary HGN presents a promising alternative, with potential advantages in interpretability and practical decision-making through its integrated graph representation.

TABLE I
PERFORMANCE METRICS OF THE ADAPTED HGN

Metric	Value
AUROC	0.7135
Accuracy	0.7578
Recall	0.8596
Precision	0.8113

V. CONCLUSION

This work introduces a novel Heterogeneous Graph Network (HGN) specifically designed for binary spatio-chemical-source prediction of PFAS contamination in groundwater. By explicitly integrating sampling locations, chemical compounds, and known contamination sources as distinct node types connected through physically meaningful relations (geographic proximity, molecular similarity, and documented source–compound associations), the proposed framework significantly outperforms traditional tabular machine learning approaches in capturing complex transport and exposure pathways.

Achieving an AUROC of 0.7135 and a recall of 0.8596 on the challenging CA-PFAS-ASGWS dataset, our model demonstrates high sensitivity for detecting contamination hotspots while providing interpretable insights into dominant contamination sources (e.g., AFFF-related sites). The Bayesian-optimized architecture ensures robust and reproducible performance, making it directly applicable for regulatory screening, prioritization of monitoring wells, and design of targeted remediation strategies.

Beyond current results, this source-aware graph paradigm opens promising avenues: incorporation of temporal dynamics through dynamic graph neural networks, integration of additional environmental covariates (soil permeability, precipitation patterns), and extension to semi-supervised learning to leverage vast unlabeled monitoring data. Ultimately, this work establishes HGNs as a powerful, interpretable alternative to black-box models in environmental contaminant modeling, with strong potential for nationwide or global-scale PFAS risk mapping.

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